The observer mass spectrum in dS_3: spacetime normalization and the BD projector

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Abstract

We distinguish fixed-mass Klein–Gordon normalization from the mass-delta normalization induced by spacetime L^2 in a specified $G = 0$ observer model. A Darboux polar decomposition gives the full radial spectrum, including its two continuum channels and bound states. The Bunch–Davies line and its antipodal image have a nonzero spacetime overlap in dS_3. We obtain the orthogonal observer projector and the resulting two-leg rescaling of the previously computed local vertex. Bounded compressed insertions provide a controlled intermediate-state identity. Group averaging, the physical state, and finite-gravity sewing remain open.

1 Spacetime L^2 and a complete radial resolution

Set the radius to one, use $G = 0$, and normalize $\int |Y_{lm}|^2 \, \mathrm{d}\Omega = 1$. For

$$ds^2 = -\,\mathrm{d}t^2 + \cosh^2 t \; \mathrm{d}\Omega_2^2, \qquad F = \sum_{lm} f_{lm}(t) Y_{lm}, \qquad \chi_{lm} = \cosh t \, f_{lm},$$

the map $R : F \mapsto (\chi_{lm})$ is unitary from spacetime L^2 to $\bigoplus_{lm} L^2(\mathbb{R}, \mathrm{d}t)$ and

$$R \Box R^{-1} = \bigoplus_{lm} (1 + H_l), \qquad H_l = -\partial_t^2 - l(l+1) \operatorname{sech}^2 t. \tag{1}$$

Each H_l is self-adjoint on $H^2(\mathbb{R})$ with core $C_c^\infty(\mathbb{R})$, since its potential is real and bounded. The full direct-sum domain additionally requires $\sum_{lm} \|(1 + H_l)\chi_{lm}\|^2 < \infty$. The finite angular-mode core fixes the closure; the larger invariant symmetric domain $C_c^\infty(dS_3)$ has the same closure. It follows that the Casimir spectral projections commute with proper, orthochronous isometries.

The reflectionless potential and its bound states are standard [1]. To fix completeness and normalization explicitly, let $D_j = \partial_t - j \tanh t$. Then

$$D_j^\dagger D_j = H_{j-1} + j^2, \quad D_j D_j^\dagger = H_j + j^2, \quad W_j = D_j (H_{j-1} + j^2)^{-1/2}. \tag{2}$$

Inductively $H_{j-1} \geq -(j-1)^2$, so the inverse square root is bounded. W_j is an isometry onto $(\ker D_j^\dagger)^\perp$, intertwines the two Hamiltonians, and the omitted kernel is spanned by $\operatorname{sech}^j t$ at energy $-j^2$. Starting with the ordinary Fourier transform of H_0 proves:

- the positive spectrum $\nu^2 > 0$ is purely absolutely continuous with two channels;
- each radial H_l has the simple eigenvalues $-k^2$, $1 \leq k \leq l$;
- the bounded zero-energy solution $P_l(\tanh t)$ is a resonance, not an L^2 eigenvector or a spectral atom.

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Thus the full $\Box$ has point eigenvalues $1-k^2$, each with infinitely many angular modes $l \geq k$. We do not identify these with a group discrete series without a separate representation analysis.

Let $d_l(\nu) = \prod_{j=1}^{l} \sqrt{\nu^2 + j^2}$. A delta-normalized continuum basis, with $\nu > 0$ for each sign, is

$$e^{\mp}_{\nu l}(t) = \frac{i^l D_l \cdots D_1 e^{\mp i\nu t}}{\sqrt{2\pi}\, d_l(\nu)}. \tag{3}$$

A real normalized bound basis is

$$b_{lk}(t) = \frac{D_l \cdots D_{k+1} \operatorname{sech}^k t}{\sqrt{B(1/2,k)} \prod_{j=k+1}^{l} \sqrt{j^2-k^2}}. \tag{4}$$

For $a_\pm(\nu) = \langle e^{\pm}_{\nu l}, \chi\rangle$ and $c_k = \langle b_{lk}, \chi\rangle$, the exact resolution is

$$\|\chi\|^2 = \int_0^\infty (|a_-|^2 + |a_+|^2)\, \mathrm{d}\nu + \sum_{k=1}^{l} |c_k|^2, \tag{5}$$

$$\chi = \int_0^\infty (e^- a_- + e^+ a_+)\, \mathrm{d}\nu + \sum_{k=1}^{l} b_{lk} c_k. \tag{6}$$

The inverse is an L^2 limit. A single continuum channel is not a resolution of the full position delta function.

2 The BD and antipodal Gram matrix

Use the KG-normalized BD mode $u_{\nu lm} = f_{\nu l} Y_{lm}$ from the preceding note [4], consistent with [2]. Set

$$q = e^{-\pi\nu}, \qquad N = [2\nu(1-q^2)]^{-1/2}, \qquad \chi_B = \cosh t\, f_{\nu l} = \sqrt{2\pi} N (e^- - q e^+).$$

The antipodal pullback $A : (t, n) \mapsto (-t, -n)$ is a self-adjoint unitary on spacetime L^2, commuting with the proper isometry group. Keeping the angular harmonic Y_{lm} fixed, its radial mode is

$$\chi_A(t) = (-1)^l \chi_B(-t) = (-1)^l \chi_B(t)^* = \sqrt{2\pi} N (e^+ - q e^-). \tag{7}$$

Complex conjugation of the full spherical harmonic also reverses m; the radial and full conjugations must be distinguished.

The continuum resolution now gives

$$\langle u^r_{\nu lm}, u^s_{\nu' l' m'} \rangle_{L^2} = G_{rs}(\nu) \delta(\nu - \nu') \delta_{ll'} \delta_{mm'}, \qquad G = \frac{\pi}{\nu} \begin{pmatrix} \coth \pi\nu & -\operatorname{csch} \pi\nu \\ -\operatorname{csch} \pi\nu & \coth \pi\nu \end{pmatrix}. \tag{8}$$

Its eigenvalues are $(\pi/\nu)\tanh(\pi\nu/2)$ and $(\pi/\nu)\coth(\pi\nu/2)$, with $\det G = (\pi/\nu)^2$. This positive spacetime Gram matrix is different from the fixed-mass KG signature $(+1,-1)$. In a basis using χ_B^* as the second radial mode, the off-diagonal entry acquires $(-1)^l$.

KL's dS_2 construction [3, eqs. (4.9)–(4.11)] has an orthogonal antipodal pair. Equation (8) is a fresh dS_3 derivation; that cross-channel orthogonality must not be transplanted between dimensions. The agreement of the diagonal normalizing function below does not imply agreement of the two Gram matrices, and is not a claim of an error in the dS_2 result.

If both nonorthogonal channels are used, the full continuous projector requires their inverse Gram matrix:

$$P_{\mathrm{ac}} = \sum_{lm} \int_0^\infty |u^r_{\nu lm}\rangle (G^{-1})_{rs} \langle u^s_{\nu lm}|\, \mathrm{d}\nu.$$

Separately normalizing each vector and adding its rank-one projector does not give the identity.

3 An explicit observer subspace and its measure

Choose and retain a mass cutoff $\nu_* > 0$. Define

$$\mu(\nu) = \frac{\nu \tanh \pi\nu}{\pi}, \qquad v_{\nu lm} = \sqrt{\mu(\nu)}\, u_{\nu lm}, \qquad (Ja)(X) = \sum_{lm} \int_{\nu_*}^{\infty} v_{\nu lm}(X) a_{lm}(\nu)\, \mathrm{d}\nu. \tag{9}$$

Then J is an isometric embedding of $\mathcal{H}_{\mathrm{obs}} = \int_{\nu_*}^{\infty} {}^{\oplus} \ell^2(l,m)\, \mathrm{d}\nu$ into spacetime L^2: $\|Ja\|^2 = \int_{\nu_*}^{\infty} \sum_{lm} |a_{lm}|^2\, \mathrm{d}\nu$. In the ordered (e^-, e^+) continuum basis,

$$v_B = \frac{(1,-q)^T}{\sqrt{1+q^2}}, \quad v_\perp = \frac{(q,1)^T}{\sqrt{1+q^2}}, \quad P_B = \frac{1}{1+q^2} \begin{pmatrix} 1 & -q \\ -q & q^2 \end{pmatrix}. \tag{10}$$

The observer projector is $1_{\nu \geq \nu_*} P_B$ on the continuum and zero on every bound mode. Consequently $J^\dagger J = 1$ and $J J^\dagger = P_{\mathrm{obs}}$, a norm-one orthogonal projection.

Both continuum channels have the same adjacent-angular-momentum boost coefficient

$$\beta_{j,m}(\nu) = \sqrt{\frac{(j^2 - m^2)(j^2 + \nu^2)}{(2j-1)(2j+1)}}. \tag{11}$$

This follows from the Darboux and spherical-harmonic recurrences, and agrees with [2, eqs. (20)–(21)] at $d = 2$, $\sigma = -1 + i\nu$. Rotations also act identically. The matrix P_B depends on ν only; hence the projection commutes with proper isometries. Equivalently, the BD space is invariant, its antipodal image is invariant, and these two lines span the continuum multiplicity space for $\nu > 0$.

In contrast to an orthogonal complement, the antipodal unit vector obeys

$$\langle v_B, A v_B \rangle = -\operatorname{sech} \pi\nu, \qquad v_\perp = \frac{A v_B + \operatorname{sech}(\pi\nu) v_B}{\tanh \pi\nu}. \tag{12}$$

The condition number of the double Gram is $\coth^2(\pi\nu/2)$. It diverges at zero, while P_B remains bounded and $\mu \sim \nu^2$. At large ν, $\mu \sim \nu/\pi$ and the antipodal overlap is exponentially small. These limits do not remove the declared cutoff.

Restoring the radius, $u_\ell = \ell^{-1/2} u_1$, $G_\ell = \ell^2 G$, and $v_\ell = \ell^{-3/2} v_1$. For physical effective mass $m = \sqrt{1+\nu^2}/\ell$,

$$\frac{\mathrm{d}\nu}{\mathrm{d}m} = \frac{\ell^2 m}{\nu}, \qquad v_m = \sqrt{\frac{\ell^2 m}{\nu}}\, v_\nu. \tag{13}$$

For $\lambda = 1 + \nu^2$, the delta-normalized mode is $v_\lambda = v_\nu/\sqrt{2\nu}$ instead. These are changes of spectral variable, not extra group Plancherel densities to multiply into $\mathrm{d}\nu$.

4 What this fixes in the local insertion

For observer slots (o, i) and a KG-normalized free-field probe p, the local coefficient of the preceding note becomes

$$\boxed{C_\eta^{\mathrm{obs}}(\nu_o, \nu_i, \nu_p) = \sqrt{\mu(\nu_o)\mu(\nu_i)}\, C_\eta^{\mathrm{KG}}(\nu_o, \nu_i, \nu_p)}. \tag{14}$$

Only the observer legs change normalization; the field's creation and annihilation operators retain their KG convention. The coefficient has dimension $\ell^{-1/2}$ with the chosen unit local coupling. This fixes the legs of this kinematic realization. It does not yet fix the earlier unknown physical two-point factor n, which also involved a state, group reduction and sewing norms.

A bounded version makes the intermediate projector unambiguous. For $g \in L^\infty(\mathrm{dS}_3)$, define

$$A_g = J^\dagger M_g J, \qquad \|A_g\| \le \|g\|_\infty, \qquad A_g^\dagger = A_{\bar{g}}. \tag{15}$$

This assignment is isometry-covariant, and

$$A_g A_h = J^\dagger M_g P_{\mathrm{obs}} M_h J, \qquad A_{gh} - A_g A_h = J^\dagger M_g (1 - P_{\mathrm{obs}}) M_h J. \tag{16}$$

For real $g = h$ the last expression is positive, being $B^\dagger B$ with $B = (1 - P_{\mathrm{obs}}) M_g J$. Replacing the intermediate spectral resolution by the full position delta would therefore change even this bounded two-point quantity.

Each fixed free-field angular mode u_p is bounded. Keeping a finite set S of such probe modes, let

$$O_S = \sum_{p \in S} (A_{u_p} \otimes a_p + A_{u_p}^\dagger \otimes a_p^\dagger).$$

Finite-particle vectors are analytic for this finite sum with bounded coefficients, so it is essentially self-adjoint on that core. Its vacuum two-point identity is

$$\langle \psi, 0 | O_S^2 | \psi, 0 \rangle = \sum_{p \in S} \|A_{u_p}^\dagger \psi\|^2 = \sum_{p \in S, lm} \int_{\nu_*}^\infty |\langle \nu l m | A_{u_p}^\dagger | \psi \rangle|^2 \, \mathrm{d}\nu. \tag{17}$$

A finite probe-mode truncation breaks full de Sitter invariance. Removing it requires independent ultraviolet and domain estimates. This kinematic identity is not a group-averaged Hartle–Hawking two-point function or a finite-G result.

5 Independent checks and the next physical step

The saved diagnostics contain 150 passed checks, with the two failed initial finite-box targets retained separately. Matrix, mode and bound state calculations use 50 decimal digits. Four $\mathrm{sech}^p t$ test functions check Parseval by actually integrating the continuous spectral weight and adding bound-state overlaps; independent time integrals check their Fourier transforms. The angular sectors include $l = 0, 1, 2, 3$, so deleting bound states produces a substantial, visible norm deficit.

A compact packet with support $0.6 \le \nu \le 1.8$ and normalized amplitude $h(\nu) \propto (\nu - 0.6)^4 (1.8 - \nu)^4$ independently checks (9) and (12). Mass quadrature uses 140/220 nodes and time quadrature 360/540 nodes. At $T = 20$, the finite-box norm deficit is 3.67314×10^{-6}, exceeding the initial 2×10^{-7} target; this failure was preserved. At $T = 40, 80$ the deficits are 5.46933×10^{-9} and 1.37811×10^{-11}. The predicted antipodal overlap is approximately -0.05055152806; it is not zero.

The missing tail estimate follows from Fourier Plancherel. For $b_\pm = h(1, -q)_\pm / \sqrt{1 + q^2}$, the first three endpoint derivatives vanish, so

$$\int_{|t|>T} |\chi(t)|^2 \, \mathrm{d}t \le T^{-8} \int_{0.6}^{1.8} (|b_-^{(4)}|^2 + |b_+^{(4)}|^2) \, \mathrm{d}\nu. \tag{18}$$

Cauchy–Schwarz bounds the omitted cross overlaps by the same coefficient. Its 64/96-node evaluations give approximately 3.26727×10^6, hence a conservative $T = 20$ estimate 1.27628×10^{-4}. The inequality is analytic; the evaluated coefficient and all finite quadratures are empirical, not interval certificates. The failed short-box tests were not erased or relabeled as a high-precision delta-normalization test.

The next task is the two-observer group-averaged inner product and its state, with an explicit Haar convention and reflection terms. Only then can this fixed local normalization be tested against the physical two-point spectrum. Independent free-probe disconnected four-point limits, removal of matter truncations, finite-G dynamics and the full holographic objective remain required.

References

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